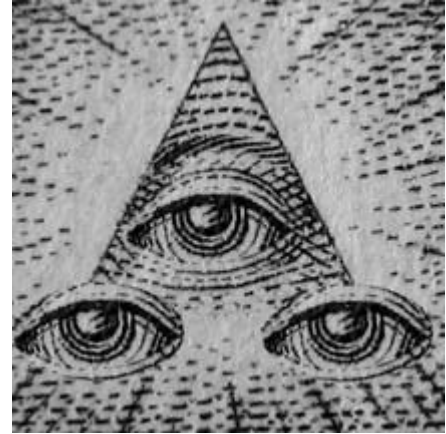


Review session Friday!
4:30-7pm in Dwinelle



Lecture 36 (Sorting 5)

Radix Sorts

CS61B, Spring 2026 @ UC Berkeley

Josh Hug and Kay Ousterhout

Sorting Stability

Lecture 36, CS61B, Spring 2026

Sorting Stability (Section Review)

Sorting Digit-by-Digit

Counting Sort

- Procedure
- Runtime

Radix Sorts

- LSD Radix Sort
- MSD Radix Sort

This week, I introduced a new topic on the discussion worksheet: Stability.

Let's review this idea since it's important for today's lecture.

- Note: Stability will be in scope for the final, e.g.
 - "Is insertion sort stable?"
 - "If you sort this array using an unstable sort, which of the following outputs are possible."
 - "Why does Java use Quicksort for ints and Merge Sort for objects?"
- The discussion worksheet provides practice with the idea of stability.

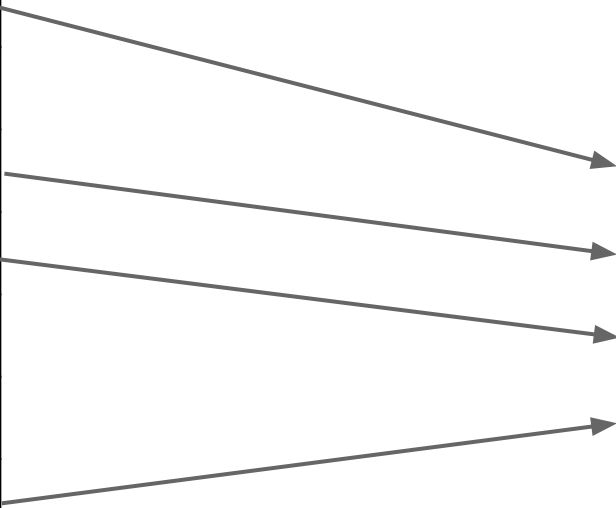
A sort is said to be stable if order of equivalent items is preserved.

sort(studentRecords, BY_NAME);

Bas	3
Fikriyya	4
Jana	3
Jouni	3
Lara	1
Nikolaj	4
Rosella	3
Sigurd	2

sort(studentRecords, BY_SECTION);

Lara	1
Sigurd	2
Bas	3
Jana	3
Jouni	3
Rosella	3
Fikriyya	4
Nikolaj	4



Equivalent items don't 'cross over' when being stably sorted.

A sort is said to be stable if order of equivalent items is preserved.

sort(studentRecords, BY_NAME);

Bas	3
Fikriyya	4
Jana	3
Jouni	3
Lara	1
Nikolaj	4
Rosella	3
Sigurd	2

sort(studentRecords, BY_SECTION);

Lara	1
Sigurd	2
Jouni	3
Rosella	3
Bas	3
Jana	3
Fikriyya	4
Nikolaj	4

Sorting instability can be really annoying! Wanted students listed alphabetically by section. 5

Sorting Digit-by-Digit

Lecture 36, CS61B, Spring 2026

Sorting Stability (Section Review)

Sorting Digit-by-Digit

Counting Sort

- Procedure
- Runtime

Radix Sorts

- LSD Radix Sort
- MSD Radix Sort

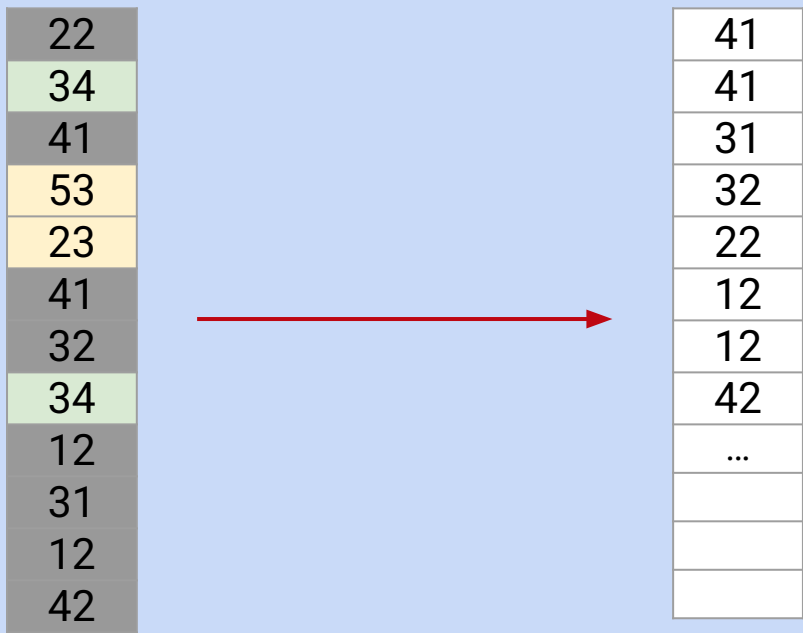
As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

- Suppose we first sort by **only the rightmost digit**.

22	41
34	41
41	31
53	32
23	22
41	12
32	12
34	42
12	...
31	
12	
42	

As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

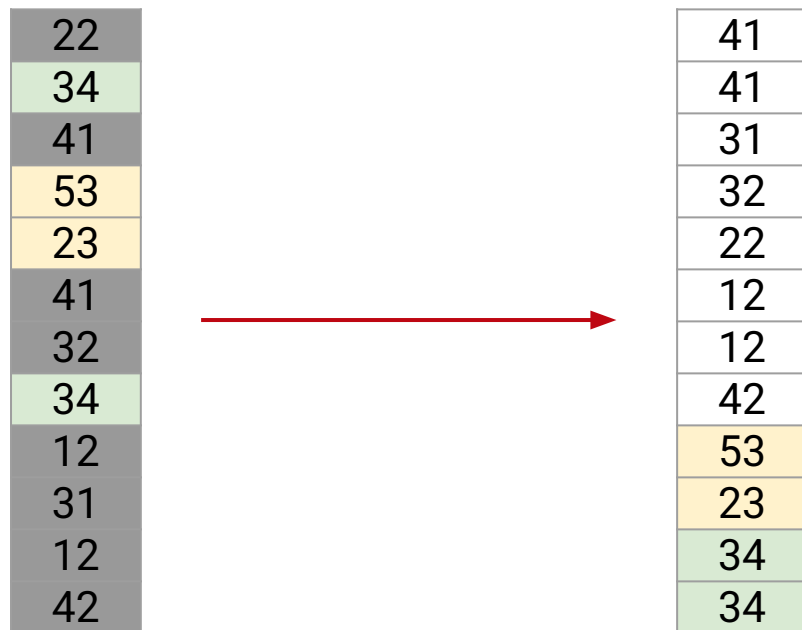
- Suppose we first sort by **only the rightmost digit**.



What are the 4 integers at the end of the array?

As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

- Suppose we first sort by **only the rightmost digit**.



As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

- Suppose we first sort by **only the rightmost digit**.

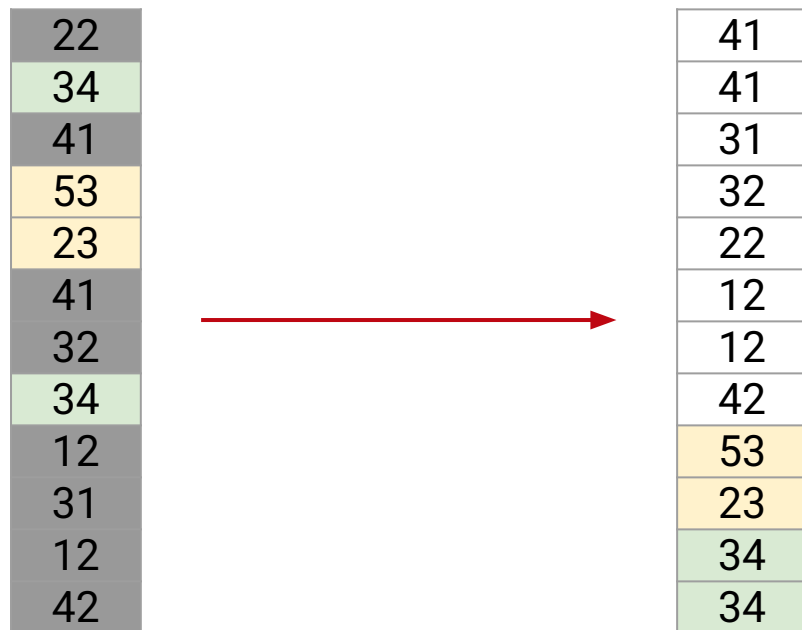
22	41
34	41
41	31
53	32
23	22
41	12
32	12
34	42
12	53
31	23
12	34
42	34

I put 53 and 23 in the order shown.
Would they always be in this order?



As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

- Suppose we first sort by **only the rightmost digit**.

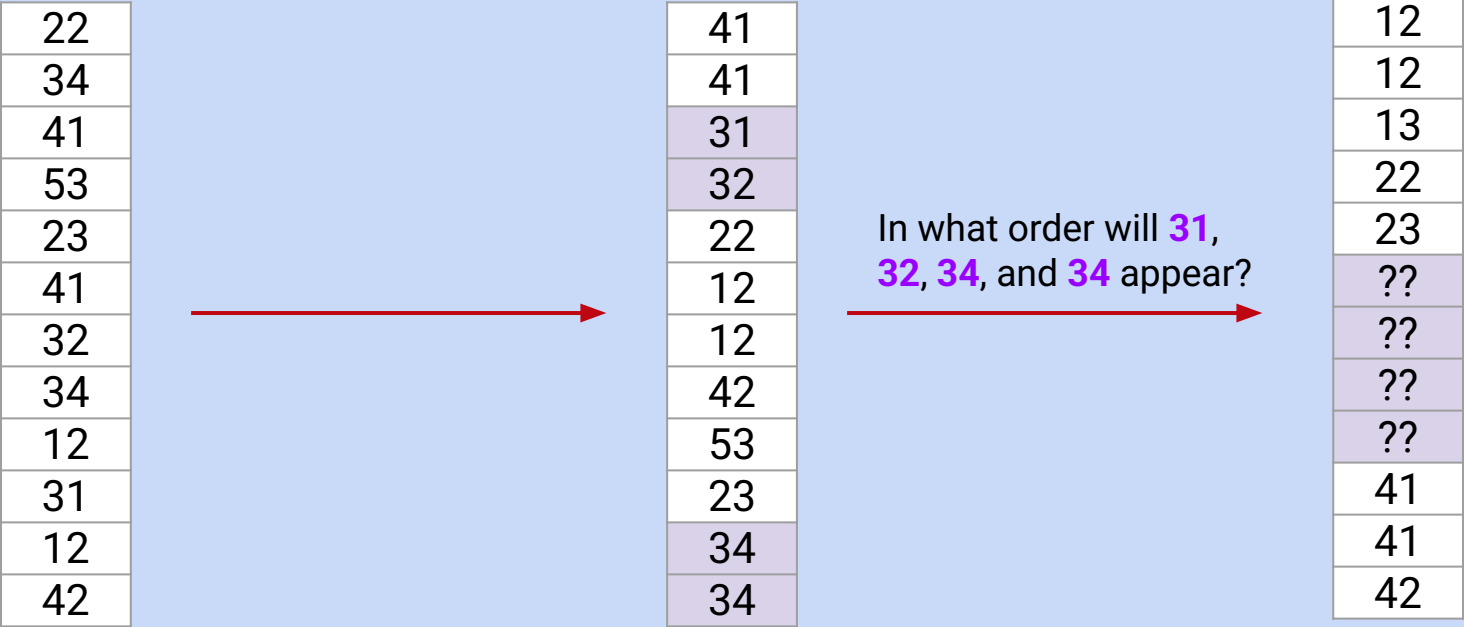


I put 53 and 23 in this order. Would they always be in this order?

- Not necessarily! Depends on if the sort I used is **stable**.
- Stable sort yields 53 then 23.
- Example: If I used Quicksort with shuffle, could have been 23 then 53.

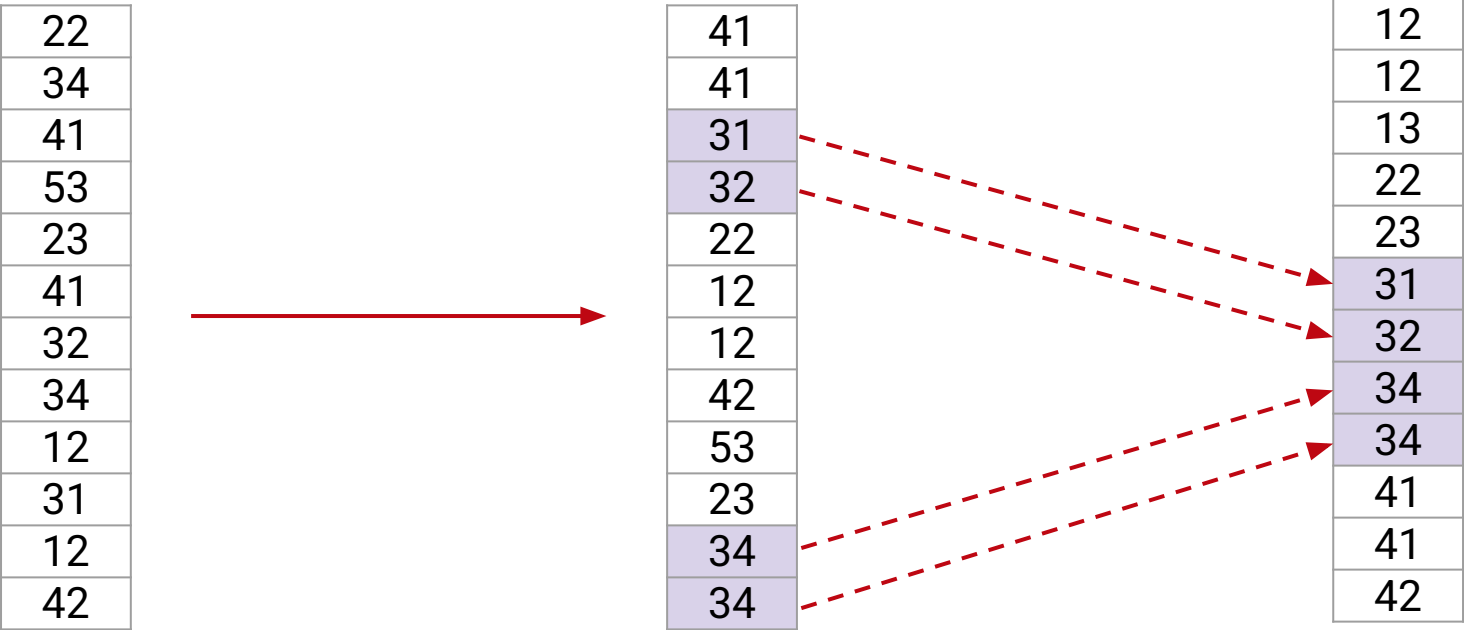
As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

- Now suppose we sort by the **left digit** using a **stable sort**.

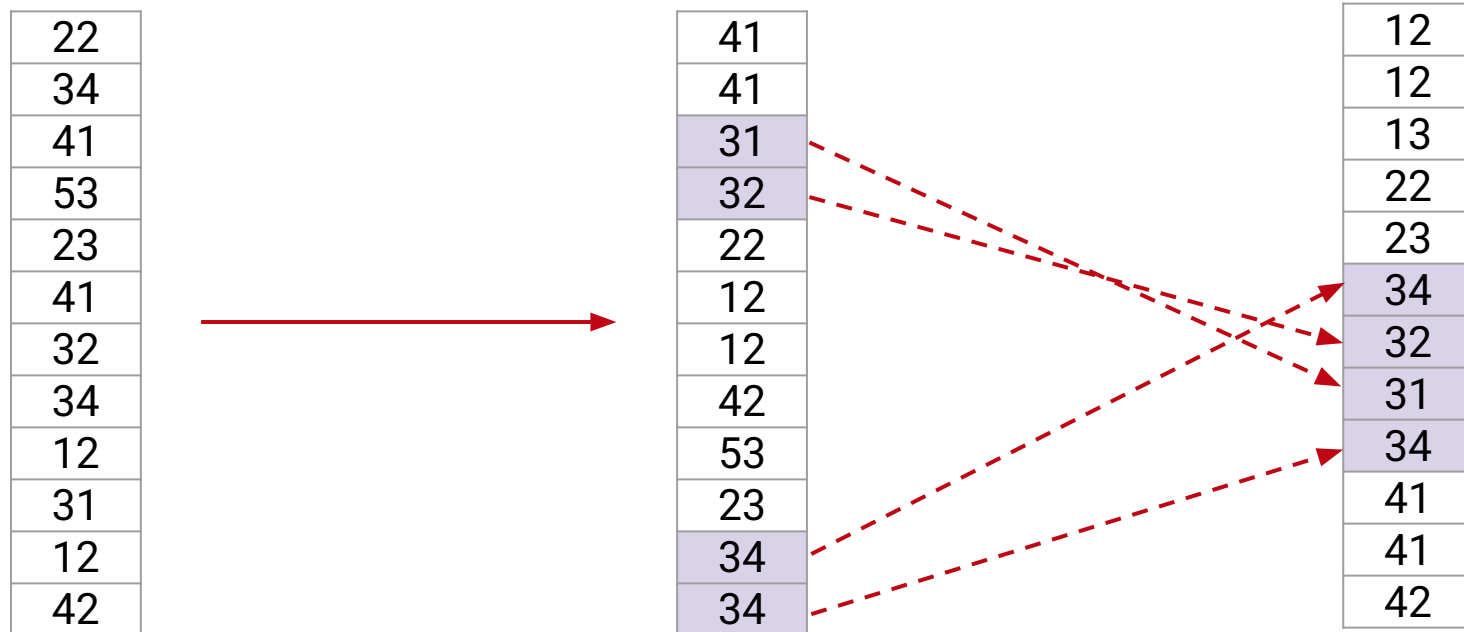


As a warmup to the later part of today's lecture. Suppose we have a list of integers we want to sort.

- Now suppose we sort by the **left digit** using a **stable sort**.

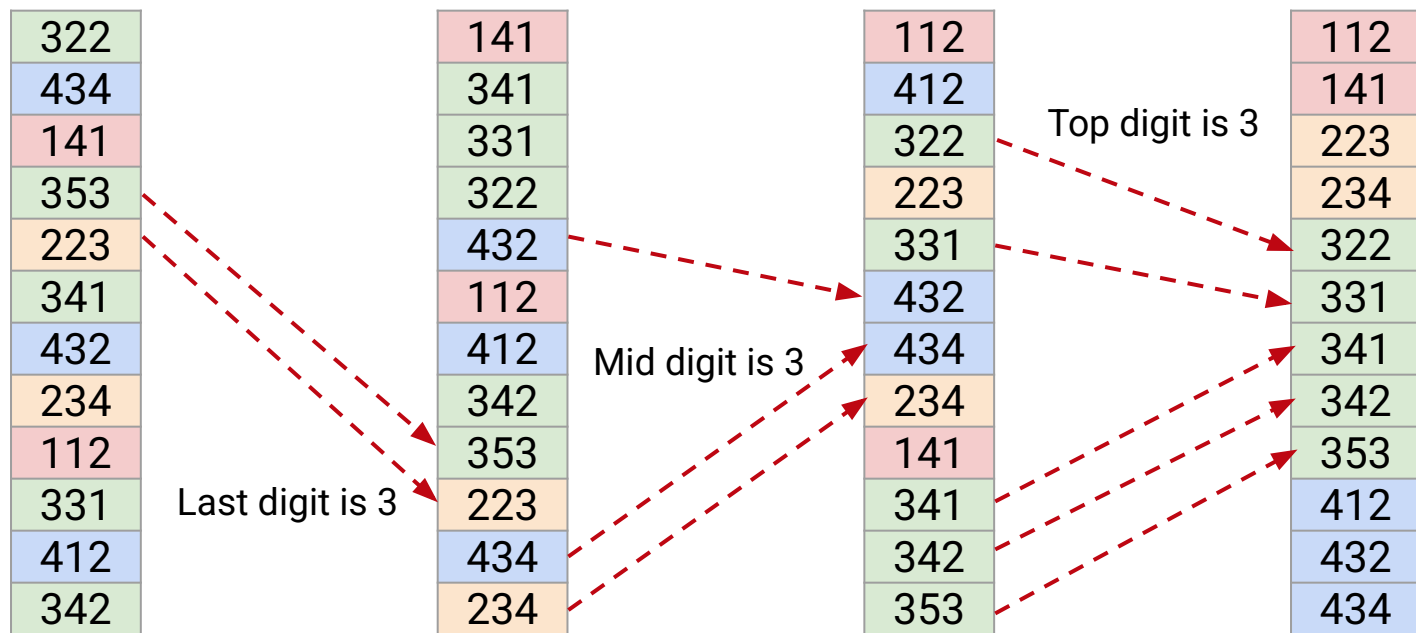


This procedure does not work if the sort subroutine is unstable.



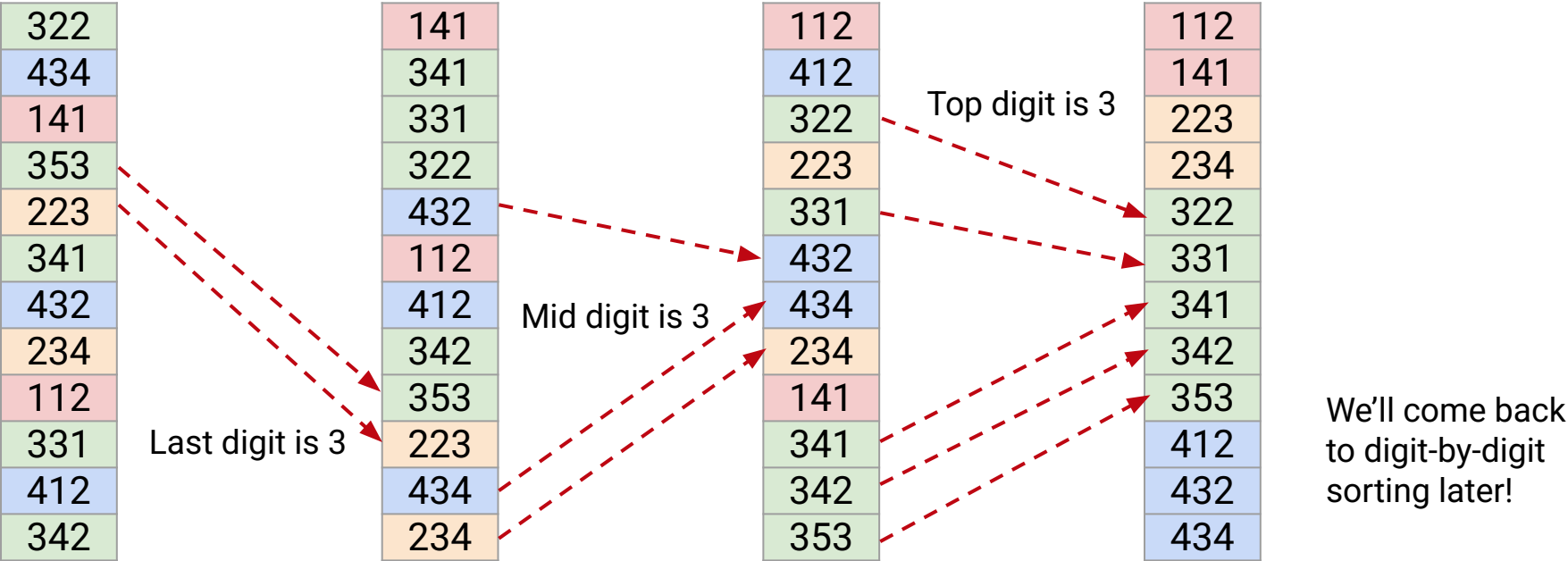
This is sometimes called “LSD sort” or “Least Significant Digit” sort.

- Select a **stable sorting** algorithm e.g. insertion sort, merge sort, etc.
- Use that stable sort on each digit, moving from **least to most significant**.
- Result of LSD sort is guaranteed to be correct (iff the sort is stable)!



Two quick notes:

- No obvious reason why this procedure is useful (can just sort by entire integer)
- Other digit-by-digit sort procedures work.



Counting Sort: Procedure

Lecture 36, CS61B, Spring 2026

Sorting Stability (Section Review)

Sorting Digit-by-Digit

Counting Sort

- **Procedure**
- Runtime

Radix Sorts

- LSD Radix Sort
- MSD Radix Sort

The key idea from our previous sorting lecture: Sorting requires $\Omega(N \log N)$ compares in the worst case.

- Thus, the ultimate comparison based sorting algorithm has a worst case runtime of $\Theta(N \log N)$.

From an asymptotic perspective, that means no matter how clever we are, we can never beat Merge Sort's worst case runtime of $\Theta(N \log N)$.

- ...but what if we don't compare at all?

Example #1: Sleep Sort (for Sorting Integers) (not actually good)

For each integer x in array A , start a new program that:

- Sleeps for x seconds.
- Prints x .

All start at the same time.

Runtime:

- $N + \max(A)$



The catch: On real machines, scheduling execution of programs must be done by an operating system. In practice requires list of running programs sorted by sleep time.

Genius sorting algorithm: Sleep sort

1 Name: **Anonymous** 2011-01-20 12:22

Man, am I a genius. Check out this sorting algorithm I just invented.

```
#!/bin/bash
function f() {
    sleep "$1"
    echo "$1"
}
while [ -n "$1" ]
do
    f "$1" &
    shift
done
wait
```

example usage:

```
./sleepsort.bash 5 3 6 3 6 3 1 4 7
```

Invented by 4chan.

Example #2: Counting Sort: Exploiting Space Instead of Time

#			
5	Sandra	Vanilla	Grimes
0	Lauren	Mint	Jon Talabot
11	Lisa	Vanilla	Blue Peter
9	Dave	Chocolate	Superpope
4	JS	Fish	The Filthy Reds
7	James	Rocky Road	Robots are Supreme
3	Edith	Vanilla	My Bloody Valentine
6	Swimp	Chocolate	Sef
1	Delbert	Strawberry	Ronald Jenkees
2	Glaser	Cardamom	Rx Nightly
8	Lee	Vanilla	La(r)va
10	Bearman	Butter Pecan	Extrobophile

Assuming keys are unique integers 0 to 11.

Idea:

- Create a new array.
- Copy item with key i into i th entry of new array.

Example #2: Counting Sort: Exploiting Space Instead of Time

#			
5	Sandra	Vanilla	Grimes
0	Lauren	Mint	Jon Talabot
11	Lisa	Vanilla	Blue Peter
9	Dave	Chocolate	Superpope
4	JS	Fish	The Filthy Reds
7	James	Rocky Road	Robots are Supreme
3	Edith	Vanilla	My Bloody Valentine
6	Swimp	Chocolate	Sef
1	Delbert	Strawberry	Ronald Jenkees
2	Glaser	Cardamom	Rx Nightly
8	Lee	Vanilla	La(r)va
10	Bearman	Butter Pecan	Extrobophile

#			
5	Sandra	Vanilla	Grimes

Example #2: Counting Sort: Exploiting Space Instead of Time

#			
5	Sandra	Vanilla	Grimes
0	Lauren	Mint	Jon Talabot
11	Lisa	Vanilla	Blue Peter
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10	Bearman	Butter Pecan	Extrobophile

#			
0	Lauren	Mint	Jon Talabot
5	Sandra	Vanilla	Grimes

Example #2: Counting Sort: Exploiting Space Instead of Time

#			
5	Sandra	Vanilla	Grimes
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5	Sandra	Vanilla	Grimes
11	Lisa	Vanilla	Blue Peter

Example #2: Counting Sort: Exploiting Space Instead of Time

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2	Glaser	Cardamom	Rx Nightly
3	Edith	Vanilla	My Bloody Valentine
4	JS	Fish	The Filthy Reds
5	Sandra	Vanilla	Grimes
6	Swimp	Chocolate	Sef
7	James	Rocky Road	Robots are Supreme
8	Lee	Vanilla	La(r)va
9	Dave	Chocolate	Superpope
10	Bearman	Butter Pecan	Extrobophile
11	Lisa	Vanilla	Blue Peter

We just sorted N items in $\Theta(N)$ worst case time.

- Avoiding yes/no questions lets us dodge our lower bound based on puppy, cat, dog!

Simplest case:

- Keys are unique integers from 0 to $N-1$.

More complex cases:

- Non-unique keys.
- Non-consecutive keys.
- Non-numerical keys.

Alphabet case: Keys belong to a finite ordered alphabet.

- Example: {♣, ♠, ♥, ♦} (in that order)

♠	Lauren
♥	Delbert
♦	Glaser
♣	Edith
♠	JS
♦	Sandra
♥	Swimp
♥	James
♣	Lee
♥	Dave
♣	Bearman
♦	Lisa

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	

Sorted



Question: What will be the index of the first ♥?

Alphabet case: Keys belong to a finite ordered alphabet.

- Example: {♣, ♠, ♥, ♦} (in that order)

♠	Lauren
♥	Delbert
♦	Glaser
♣	Edith
♠	JS
♦	Sandra
♥	Swimp
♥	James
♣	Lee
♥	Dave
♣	Bearman
♦	Lisa

0	♣	
1	♣	
2	♣	
3	♠	
4	♠	
5		
6		
7		
8		
9		
10		
11		

Sorted

Question: What will be the index of the first ♥?

Counting sort:

- Count number of occurrences of each item.
- Iterate through list, using count array to decide where to put everything.
- Slide [Demo](#)

Bottom line, we can use counting sort to sort N objects in $\Theta(N)$ time.

Counting Sort: Runtime

Lecture 36, CS61B, Spring 2026

Sorting Stability (Section Review)

Sorting Digit-by-Digit

Counting Sort

- Procedure
- **Runtime**

Radix Sorts

- LSD Radix Sort
- MSD Radix Sort

For sorting an array of the 100 largest cities by population, which sort do you think has a better expected worst case runtime in seconds?

- A. Counting Sort (as described in our demo)
- B. Quicksort

First question to ask yourself: What is the alphabet for counting sort here?



Example input:

Population	City Name
800000	San Francisco
12000	Seabrook

For sorting an array of the 100 largest cities by population, which sort do you think has a better expected worst case runtime in seconds?

- A. Counting Sort (as described in our demo)
 - a. The alphabet for counting sort is $\{1, 2, 3, \dots, 36953600\}$
- B. Quicksort

Counting sort requires building an array of size 36,953,600 (population of Tokyo).

9272670	Ahmedabad
5921200	Alexandria
5618890	Ankara
6482182	Atlanta
8500000	Bandung
14771700	Bangalore
...	...



...	...
4777999	0
4778000	1
4778001	0
4778002	0
...	...
36953600	1



...

Counts

What is the runtime for counting sort on N keys with alphabet of size R ?

- Treat R as a variable, not a constant.

This is a tough question!

The slide [demo](#) might be helpful.



Counting Sort Runtime Analysis

Total runtime on N keys with alphabet of size R : $\Theta(N+R)$

- Creating and filling our count-related arrays: $\Theta(R)$
 - Example: $R = 4$ for four card suits.
- Counting each item and copying into new array: $\Theta(N)$

For ordered array.

For counts and starting points.

Memory usage: $\Theta(N+R)$

Empirical experiments needed to compare vs. Quicksort on practical inputs.

Bottom line: If N is $\geq R$, then we expect reasonable performance.

Total runtime on N keys with alphabet of size R : $\Theta(N+R)$

- Create an array of size R to store counts: $\Theta(R)$
- Counting number of each item: $\Theta(N)$
- Calculating target positions of each item: $\Theta(R)$
- Creating an array of size N to store ordered data: $\Theta(N)$
- Copying items from original array to ordered array: Do N times:
 - Check target position: $\Theta(1)$
 - Update target position: $\Theta(1)$
- Copying items from ordered array back to original array: $\Theta(N)$

For ordered array.

For counts and starting points.

Memory usage: $\Theta(N+R)$

Empirical experiments needed to compare vs. Quicksort on practical inputs.

Bottom line: If N is $\geq R$, then we expect reasonable performance.

Give an example of a specific situation where Counting Sort will be clearly faster than Quicksort.

- Counting Sort: $\Theta(N+R)$
- Quicksort: $\Theta(N \log N)$

Previous example was sorting $N = 100$ cities by population ($R = 37,832,892$).



	Memory	Runtime	Notes	Stable?
Heapsort	$\Theta(1)$	$\Theta(N \log N)$	Bad caching (61C)	No
Insertion	$\Theta(1)$	$\Theta(N^2)$	Small N, almost sorted	Yes
Mergesort	$\Theta(N)$	$\Theta(N \log N)$	Fastest stable	Yes
Random Quicksort	$\Theta(\log N)$	$\Theta(N \log N)$ expected	Fastest compare sort	No
Counting Sort	$\Theta(N+R)$	$\Theta(N+R)$	Alphabet keys only	Yes

N: Number of keys. R: Size of alphabet.

Counting sort is nice, but alphabetic restriction limits usefulness.

- Idea: Let's try digit-by-digit sorting.
- The set of possible digits will be a relatively small alphabet.

LSD Radix Sort

Lecture 36, CS61B, Spring 2026

Sorting Stability (Section Review)

Sorting Digit-by-Digit

Counting Sort

- Procedure
- Runtime

Radix Sorts

- **LSD Radix Sort**
- MSD Radix Sort

Counting sort is slow when the alphabet is large.

- By decomposing input into a string of characters from a finite alphabet, we can force R to be small.

horse	Lauren
elf	Delbert
cat	Glaser
crab	Edith
monkey	JS
rhino	Sandra
raccoon	Swimp
cat	James
fish	Lee
tree	Dave
virus	Bearman
human	Lisa

♠♠	Lauren
♥♦	Delbert
♦♣	Glaser
♣♥	Edith
♠♥	JS
♦♣	Sandra
♥♠	Swimp
♥♦	James
♣♠	Lee
♥♣	Dave
♣♠	Bearman
♦♠	Lisa

4238	Lauren
34163	Delbert
123	Glaser
43415	Edith
9918	JS
767	Sandra
3	Swimp
634	James
724	Lee
2346	Dave
457	Bearman
312	Lisa

Digit by Digit Sorting (Redux)

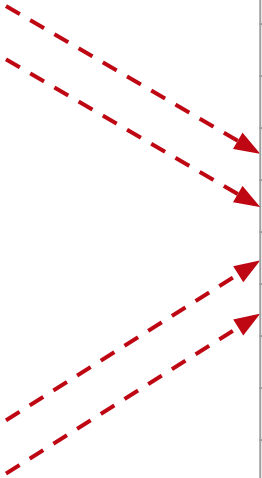
As we've seen, we can sort each digit independently from rightmost digit towards left.

- Example over the alphabet {♣, ♠, ♥, ♦}

♠♠	Lauren
♥♦	Delbert
♦♣	Glaser
♣♥	Edith
♠♥	JS
♦♣	Sandra
♥♠	Swimp
♥♦	James
♣♠	Lee
♥♣	Dave
♣♠	Bearman
♦♠	Lisa



♦♣	Glaser
♦♣	Sandra
♥♣	Dave
♥♠	Swimp
♠♠	Lauren
♣♠	Lee
♣♠	Bearman
♦♠	Lisa
♠♥	JS
♣♥	Edith
♥♦	James
♥♦	Delbert

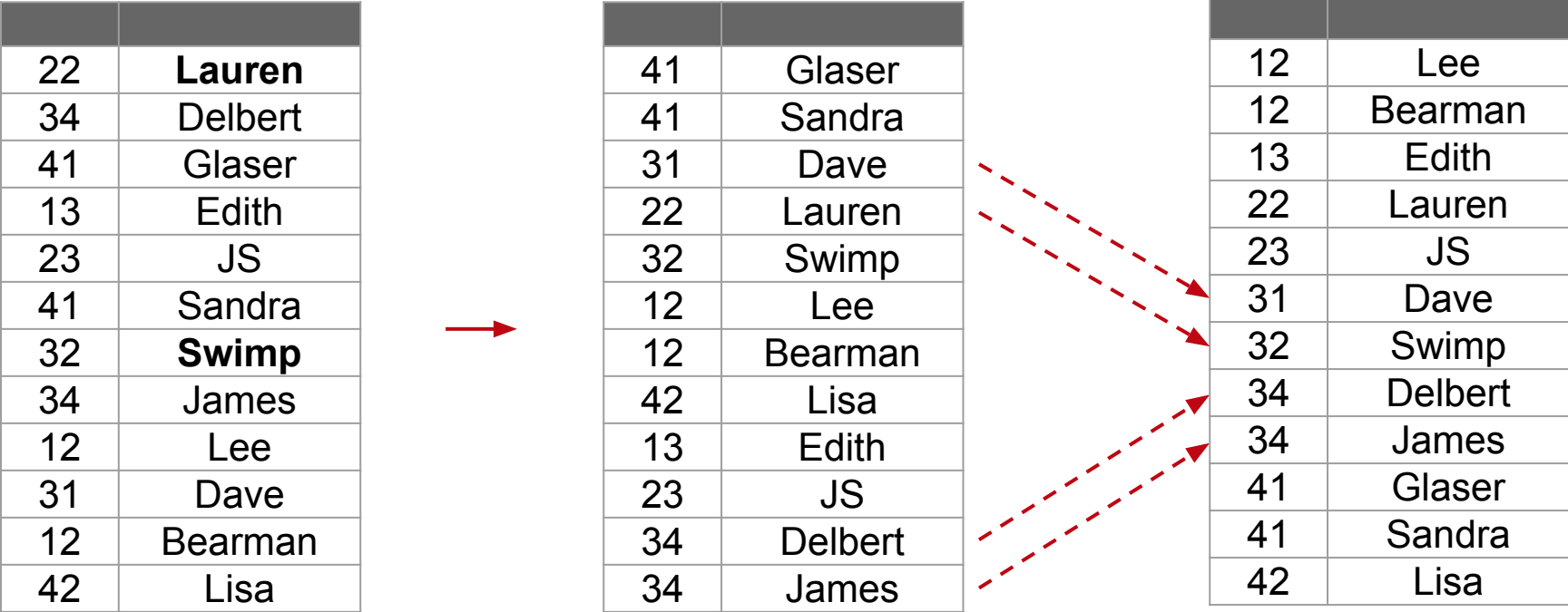


♣♠	Lee
♣♠	Bearman
♣♥	Edith
♠♠	Lauren
♠♥	JS
♥♣	Dave
♥♠	Swimp
♥♦	James
♥♦	Delbert
♦♣	Glaser
♦♣	Sandra
♦♠	Lisa

LSD (Least Significant Digit) Radix Sort -- Using Counting Sort

As we've seen, we can sort each digit independently from rightmost digit towards left.

- Example over the alphabet {1, 2, 3, 4}



Non-comparison based sorting algorithms that proceed digit-by-digit are called “Radix Sorts”.

Via wikipedia: “In a positional numeral system, the **radix** or base is the number of unique digits, including the digit zero, used to represent numbers.”

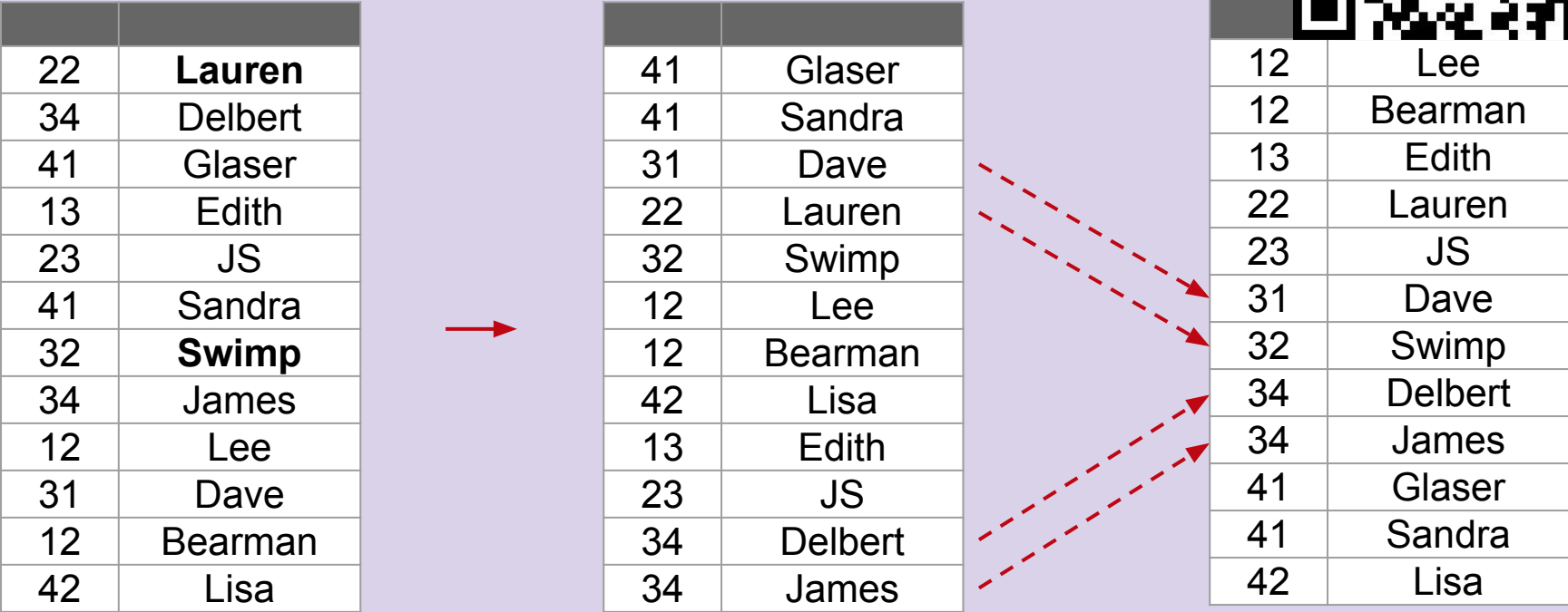
The sort we’ve just discussed is called “LSD Radix Sort”.

- LSD: Least Significant Digit.



What is the runtime of LSD Radix sort?

- Pick appropriate letters to represent non-constant terms.
- Recall runtime for counting sort was $\Theta(N+R)$.



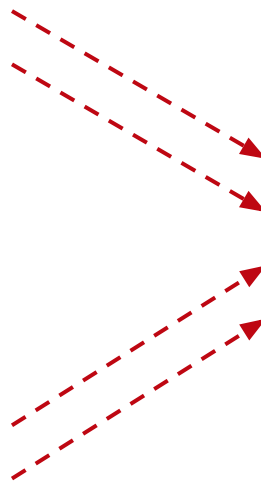
What is the runtime of LSD sort?

- $\Theta(WN+WR)$
- N: Number of items, R: size of alphabet, W: Width of each item in # digits

22	Lauren
34	Delbert
41	Glaser
13	Edith
23	JS
41	Sandra
32	Swimp
34	James
12	Lee
31	Dave
12	Bearman
42	Lisa



41	Glaser
41	Sandra
31	Dave
22	Lauren
32	Swimp
12	Lee
12	Bearman
42	Lisa
13	Edith
23	JS
34	Delbert
34	James



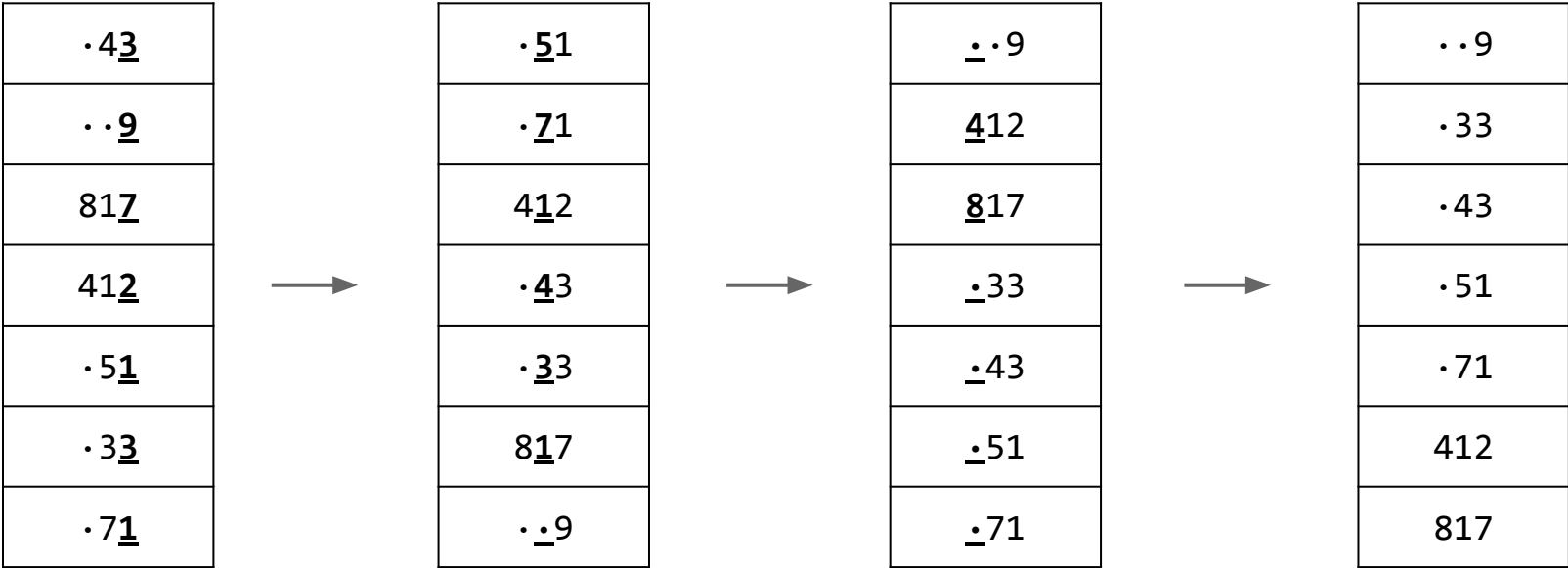
12	Lee
12	Bearman
13	Edith
22	Lauren
23	JS
31	Dave
32	Swimp
34	Delbert
34	James
41	Glaser
41	Sandra
42	Lisa

After processing least significant digit, we have array shown below. Now what?

4 <u>3</u>	→	51
<u>9</u>		71
81 <u>7</u>		412
41 <u>2</u>		43
5 <u>1</u>		33
3 <u>3</u>		817
7 <u>1</u>		9

Non-equal Key Lengths

When keys are of different lengths, can treat empty spaces as less than all other characters.



W passes of counting sort: $\Theta(WN+WR)$ runtime.

- Annoying feature: Runtime depends on length of longest key.

	Memory	Runtime	Notes	Stable?
Heapsort	$\Theta(1)$	$\Theta(N \log N)^*$	Bad caching (61C)	No
Insertion	$\Theta(1)$	$\Theta(N^2)^*$	Small N, almost sorted	Yes
Mergesort	$\Theta(N)$	$\Theta(N \log N)^*$	Fastest stable sort	Yes
Random Quicksort	$\Theta(\log N)$	$\Theta(N \log N)^*$ expected	Fastest compare sort	No
Counting Sort	$\Theta(N+R)$	$\Theta(N+R)$	Alphabet keys only	Yes
LSD Sort	$\Theta(N+R)$	$\Theta(WN+WR)$	Strings of alphabetical keys only	Yes

N: Number of keys. R: Size of alphabet. W: Width of longest key.

*: Assumes constant compareTo time.

MSD Radix Sort

Lecture 36, CS61B, Spring 2026

Sorting Stability (Section Review)

Sorting Digit-by-Digit

Counting Sort

- Procedure
- Runtime

Radix Sorts

- LSD Radix Sort
- **MSD Radix Sort**

Basic idea: Just like LSD, but sort from leftmost digit towards the right.

Pseudopseudohypoparathyroidism
Floccinaucinihilipilification
Antidisestablishmentarianism
Honorificabilitudinitatibus
Pneumonoultramicroscopicsilicovolcanoconiosis

Suppose we sort by topmost digit, then middle digit, then rightmost digit. Will we arrive at the correct result? A. Yes, B. No

a	d	d
c	a	b
f	a	d
f	e	e
b	a	d
b	e	e
f	e	d
b	e	d
a	c	e



a	d	d
a	c	e
b	a	d
b	e	e
b	e	d
c	a	b
f	a	d
f	e	e
f	e	d

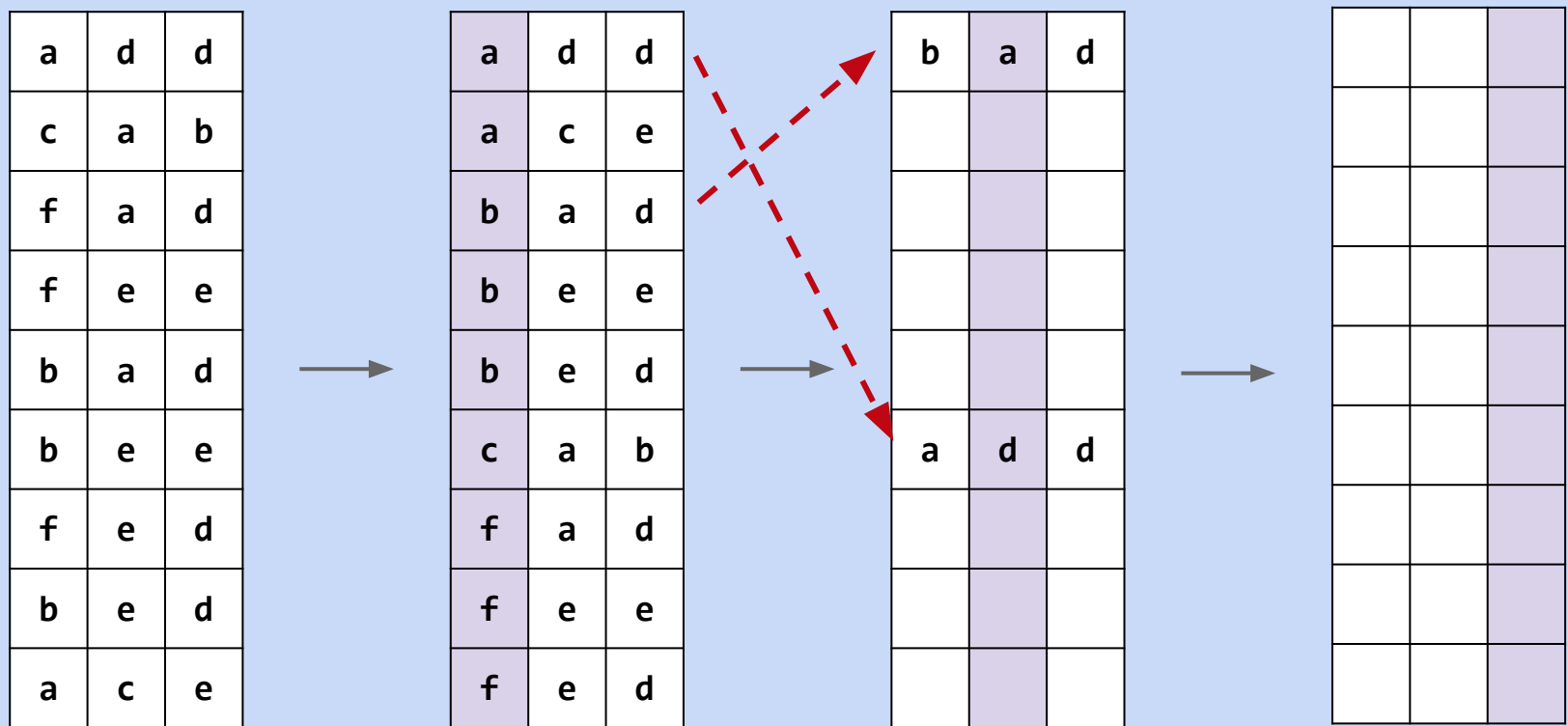




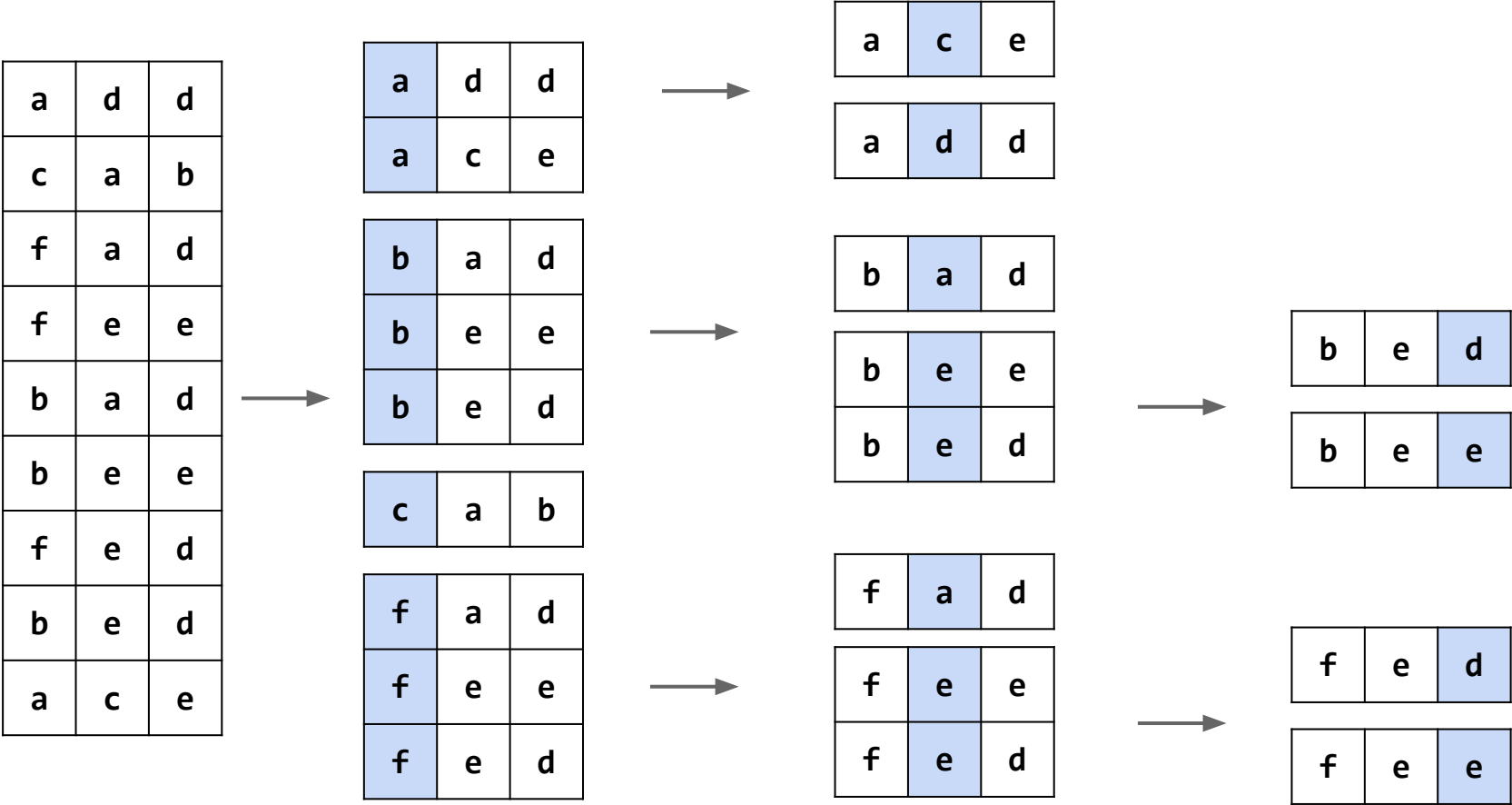


MSD Sort Question

Suppose we sort by topmost digit, then middle digit, then rightmost digit. Will we arrive at the correct result? A. Yes, **B. No.** How do we fix?



Key idea: Sort each subproblem separately.



What is the Best Case of MSD sort (in terms of N , W , R)?

What is the Worst Case of MSD sort (in terms of N , W , R)?

Again, recall counting sort is $\Theta(N + R)$.

- And LSD radix sort was just W counting sorts, so it was $WN + WR$.

No Poll

Best Case.

- We finish in one counting sort pass, looking only at the top digit: $\Theta(N + R)$

Worst Case.

- We have to look at every character, degenerating to LSD sort: $\Theta(WN + WR)$
- In other words, we do W counting sorts.

Sorting Runtime Analysis

	Memory	Runtime (worst)	Notes	Stable?
Heapsort	$\Theta(1)$	$\Theta(N \log N)^*$	Bad caching (61C)	No
Insertion	$\Theta(1)$	$\Theta(N^2)^*$	Fastest for small N, almost sorted data	Yes
Mergesort	$\Theta(N)$	$\Theta(N \log N)^*$	Fastest stable sort	Yes
Random Quicksort	$\Theta(\log N)$	$\Theta(N \log N)^*$ expected	Fastest compare sort	No
Counting Sort	$\Theta(N+R)$	$\Theta(N+R)$	Alphabet keys only	Yes
LSD Sort	$\Theta(N+R)$	$\Theta(WN+WR)$	Strings of alphabetical keys only	Yes
MSD Sort	$\Theta(N+WR)$	$\Theta(N+R)$ (best) $\Theta(WN+WR)$ (worst)	Bad caching (61C)	Yes

N: Number of keys. R: Size of alphabet. W: Width of longest key.

*: Assumes constant compareTo time.

Why did you choose to enroll (or not) in language courses at UC Berkeley?

Learn more:



Survey link:



Your participation in this 10-20 minute
survey will help advance understanding
of undergraduate needs.

This ACTFL survey is
supported by the Berkeley
Language Center



Sounds of Sorting Algorithms

Starts with selection sort: <https://www.youtube.com/watch?v=kPRA0W1kECg>

Insertion sort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=0m9s>

Quicksort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=0m38s>

Mergesort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=1m05s>

Heapsort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=1m28s>

LSD sort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=1m54s>

MSD sort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=2m10s>

Shell's sort: <https://www.youtube.com/watch?v=kPRA0W1kECg&t=3m37s>

Questions to ponder (later... after class):

- How many items are sorted in the video for selection sort?
- Why does insertion sort take longer / more compares than selection sort?
- At what time stamp does the first partition complete for Quicksort?
- Could the size of the input used by mergesort in the video be a power of 2?
- What do the colors mean for heapsort?
- How many characters are in the alphabet used for the LSD sort problem?
- How many digits are in the keys used for the LSD sort problem?

Creepy eye thing, title slide:

http://photos3.meetupstatic.com/photos/event/a/3/f/4/highres_335141972.jpeg